



Electrochemical performance of supercapacitor electrodes based on carbon aerogel-reinforced spread tow carbon fiber fabrics

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ABSTRACT

Fabric-based supercapacitor electrodes were fabricated by embedding spread tow carbon fiber fabrics, in monolithic, bicontinuous carbon aerogels (CAG). The incorporation of CAG, at less than 30 wt%, increased the specific surface area of the CAG-CF fabric to above $230 \text{ m}^2 \text{ g}^{-1}$ and the pore volume to about $0.35 \text{ cm}^3 \text{ g}^{-1}$, orders of magnitude higher than that for the as-received carbon fibres. The presence of the CAG not only improves the electrochemical performance of the composite electrodes but may enhance the mechanical response due to the high stiffness of the aerogel structure. Cyclic voltammetry, galvanostatic charge-discharge and electrochemical impedance measurements were performed on symmetric supercapacitor cells consisting of two CAG-reinforced fabrics in an ionic liquid electrolyte. The specific capacitance of the symmetric supercapacitor was determined to be in the range $3\text{--}5 \text{ F g}^{-1}$, considerably higher than that for the plain carbon fibres. Since optimum structural electrolytes are not yet available, this value was normalized to the total mass of both electrodes to place an upper bound on future structural supercapacitors using this spread tow CAG-CF system. The maximum specific energy and specific power, normalized to the total mass of the electrodes, were around 2.64 Wh kg^{-1} and 0.44 kW kg^{-1} , respectively. These performance metrics demonstrate that the thin CAG-modified spread tow fabrics are promising electrodes for future use in structural supercapacitors. In principle, in future devices, the reduced ply thickness offers both improved mechanical properties and shorter ion diffusion distance, as well as opportunities to fabricate higher voltage multicell assemblies within a given component geometry.

1. Introduction

Structural supercapacitors combine electrochemical energy storage with structural integrity [1–6]. So far, two approaches have been proposed for the production of structural power composites: (a) embedding of conventional electrochemical devices into structural composites particularly carbon fiber-reinforced plastics [7–9], and (b) a more integrated, multifunctional materials approach in which the structural constituent functions simultaneously as an energy storing electrode in addition to providing mechanical, load-bearing functionality [10–15]. The latter option, in principle, offers greater weight-saving if the conflicting materials demands can be resolved. Multifunctional devices based on supercapacitors are attractive for their high cycle life, dimensional stability, and relative environmental tolerance (for example, when compared to lithium-ion cells). Supercapacitors, also

known as electrochemical double layer capacitors, offer high power density, though lower energy density, than battery systems; hence, they play important roles in power systems for consumer electronics and electric vehicles [16]. One promising structural supercapacitor embodiment exploits laminates of carbon fiber fabrics sandwiching an insulating fabric separator. Embedding the carbon fibers within a bicontinuous carbon aerogel matrix not only provides a high surface area electrode, but also enhances the mechanical performance [10,14].

Carbon fiber fabrics offer the required mechanical performance, but have low specific surface area which means the energy stored in the electric double layer is insufficient, with capacitances much less than 1 F g^{-1} [17,18]. Traditional activated (non-structural) carbon fibres are prepared from carbonized polymer precursors, so have inadequate mechanical or electrical performance. Surface activation of structural carbon fibers can improve double layer capacity [19–23], and may also

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introduce redox-active functional groups that add to the effective capacitance [19,21,24–26]. However, the activation process must be carefully controlled to avoid degrading the electrical conductivity and the mechanical performance of the fibers. In any case, the process is confined to the surface of the fibres, limiting the overall specific electrochemical active surface area. As an alternative, greater energy storage capability can be achieved by exploiting the spaces between the primary reinforcing carbon fibres. High surface area carbon or redox-active materials can be infiltrated into the inter-fiber voids or deposited onto the surface of the carbon fibers [17,24,25,27–29]. One particularly promising option is to fill the large proportion of the matrix volume with a continuous carbon aerogel. In particulate form, deposited by slurry casting, such aerogels are widely used in conventional electrochemical capacitors. However, in the presence of carbon fibres, it is possible to create stiff, continuous monoliths that can also contribute mechanically [2,10,12]. Such monolithic electrode designs can enhance electrical percolation and hence conductivity, in turn, maximizing power performance in structural supercapacitors. In principle, thinner plies of carbon fiber fabrics are advantageous for structural supercapacitors, from both the structural and electrochemical perspective. Mechanically, thin ply composites are attracting considerable attention for their higher strength, stiffness, and delamination resistance compared to conventional plies. The reduced ply thickness lowers the interlaminar shear stresses which develop at the interfaces, and the reduced crimp improves alignment and volume fraction [30]. Electrochemically, thinner plies are beneficial as they have shorter ionic conduction paths, reducing the voltage drop within the electrodes and improving power density. In addition, the thinner plies will allow more cells to be assembled within a given thickness (often pre-defined by the application or component geometry), enabling multicell assemblies, attractive for many applications that operate at higher voltage [31].

In this work, we introduce carbon fiber spread tow fabrics [30,32], manufactured using state-of-the-art fiber spreading technology, as scaffolds for carbon aerogels which were then used as electrodes in symmetric supercapacitors. The fabrics were embedded within carbon aerogels using a Resin Infusion Under Flexible Tooling (RIFT) approach [10,33] to infuse resorcinol-formaldehyde precursors, followed by thermal curing and pyrolysis at elevated temperature. Physical characterization indicates that the aerogel-reinforced materials exhibit high specific surface areas and pore volumes, as required for applications in supercapacitors. To demonstrate the suitability of the materials as electrodes in supercapacitors, symmetric cells were constructed and characterized using cyclic voltammetry, galvanostatic cycling and electrochemical impedance tests. The electrochemical data were used to evaluate the performance metrics (specific capacitance, specific energy, specific power, equivalent series resistance, and relaxation time constant) and compared to other related supercapacitors.

2. Materials and Methods

2.1. Materials and synthesis

All chemicals were used as received. The resin precursor used to manufacture the carbon aerogels was developed via base-promoted condensation reaction of resorcinol with formaldehyde [34–36]. Typically, 50 g of formaldehyde (615 mmol, 37 % wt. In H₂O, Sigma-Aldrich®) and 34 g of resorcinol (309 mmol, ≥99.0%, Sigma-Aldrich®) were mixed in 47 g deionised water. After adding 0.35 g of KOH (6.2 mmol) to initiate the polymerisation reaction, as shown by the scheme in Fig. S1, the resulting mixture was left stirring at ambient temperature for about 3 h. It was then transferred to a water bath kept at 30 °C and allowed to react under constant stirring for about 3 h. The reddish resorcinol-formaldehyde (RF) solution ('resol') was stored in refrigerator at 5 °C for 24–40 h until the subsequent resin infusion process. A pictorial summary of the resin development is provided in Fig. S2. The chilling step was needed to slow down the reactions (polycondensation

and Ostwald ripening) and increase the polymer content in the resin before infusion into the carbon fiber fabrics. Resin Infusion under Flexible Tooling (RIFT) was adopted to impregnate the carbon fiber fabrics (10 cm × 10 cm) with the resin [10,33]. As shown in Fig. 1, the RF-infused carbon fiber fabrics (TeXtreme® 0/90° 43 PW HS40 W0/20:21/20:21–1000, 43 g m⁻² spread tow carbon fiber fabrics from OXEON® AB in Sweden) were wrapped within vacuum bagging plastic, infused and then cured at room temperature over 24 h and subsequently transferred to an oven for further reaction at 50 and 80 °C for a total of 48 h. The fully cured RF-carbon fiber composite laminates were then immersed in an acetone bath to exchange for water trapped in the pores and then dried at ambient temperature. A schematic synopsis of the steps is shown in Fig. S3. The laminates were subsequently carbonized in a box furnace (KSL-1700X-H2 from MTI) at 800 °C for 1 h under a continuous flow of nitrogen (0.3 L min⁻¹) to obtain carbon aerogel-reinforced carbon fiber fabrics. The CAG mass loading was calculated as follows: a control fabric with 15 wt % sizing was pyrolyzed at the same temperature and used as a reference to establish the mass loading of the CAG in the CAG-CF hybrid fabric as shown below.

$$m(\text{CAG}) = m(\text{CAG} + \text{CF})_{\text{pyrolysis}} - m(\text{CF})_{\text{pyrolysis}} \quad (1)$$

2.2. Characterization

2.2.1. Thermogravimetry

Thermogravimetric analyses of RF-infused carbon fiber fabrics were conducted using Mettler Toledo TGA/DSC under a continuous flow of N₂ gas (60 mL min⁻¹). In a typical experiment, 4.5 mg RF powder detached from the fabrics was placed in alumina pans (70 µL volume) and heated at a rate of 5 °C min⁻¹ to 800 °C and held at that temperature for 1 h.

2.2.2. Scanning electron microscopy (SEM)

Samples of CAG-CF were mounted on Al-stubs using carbon and silver tapes and imaged in a Zeiss Auriga SEM, using secondary electrons. A working distance of 5 mm and acceleration voltage of 5 kV were used for high resolution imaging with the Inlens detector. Low magnification imaging was performed at a working distance of 40 mm and at an acceleration voltage of 10 kV, using the SESI detector.

2.2.3. Nitrogen gas physisorption: surface area and porosity analysis

Prior to analysis, the samples were dried at 250 °C under continuous flow of N₂ gas for 24 h. Afterwards, the samples in the glass sample tubes were weighed and mounted on the analysis ports of a Tri Star II analyzer from Micromeritics. The analyses were carried out at 77 K in a liquid N₂ bath. The free space for each sample was evaluated under same conditions using He gas to account for the dead volume of the sample tubes. To evaluate the specific surface areas, the BET model [37] was applied in the relative pressure (p/p₀) range of 0.05–0.3. The pore width distributions were extracted from the isotherms using non-local density functional theory (NLDFT) models with the assumption of slit-shaped pores [38].

2.2.4. Assembling symmetric supercapacitors and electrochemical testing

The electrochemical performance of the CAG-CF fabrics was studied using cyclic voltammetry (CV) and galvanostatic charge-discharge (GCD) techniques. Pouch cells were constructed using two CAG-CF fabrics (4 cm × 4 cm × 107 ± 4 µm, total mass ≈ 100 mg) as negative and positive electrodes in ionic liquid electrolyte, 1-ethyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide (EMI-TFSI) (2–5 mL), sealed in vacuum bags under reduced pressure (200 mbar or 0.197 atm). Non-woven glass fiber fabrics (6 cm × 6 cm) with a thickness of ~240 µm (30 g m⁻², 20103 A, Optiveil® E-glass from Technical Fiber Products) were used as separators between the two symmetric electrodes. The cells were fabricated outside the glovebox, and thus there might be some risk of contamination with moisture from the ambient atmosphere. Strips of graphite foil were placed on each electrode and used as tabs for

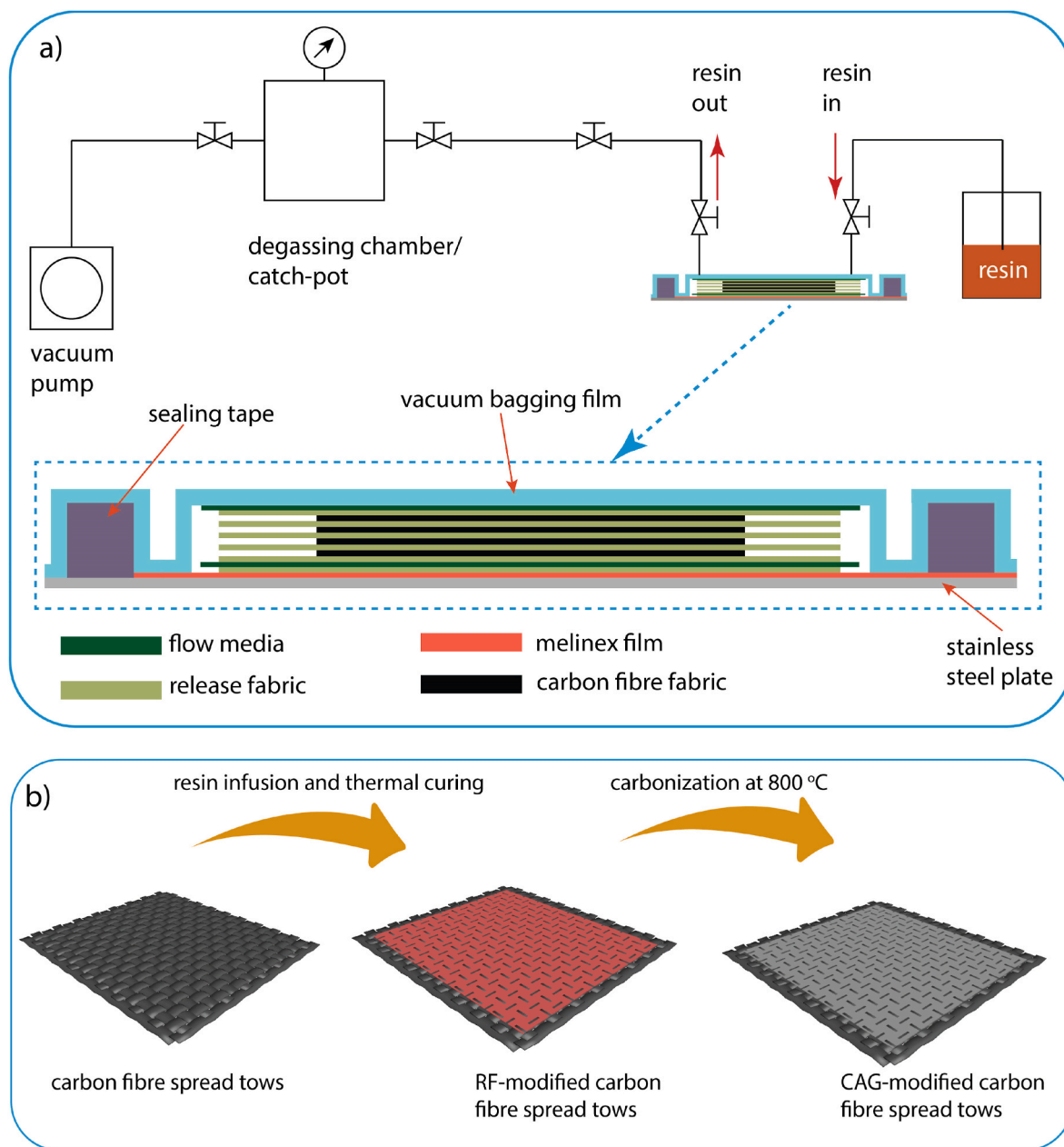


Fig. 1. Schematic presentation of the manufacturing process for carbon aerogel-modified carbon fiber laminates (a) A typical RIFT setup used to impregnate the spread tow carbon fibre fabrics with resorcinol-formaldehyde resin. (b) The resin infused into the fabrics is cured at ambient temperature for 24 h and then crosslinked at 50 °C (24 h) and 80 °C (24 h) to prepare organic aerogels, which are eventually carbonized at 800 °C.

electrical contact. Measurements were done using SP240 potentiostat (Bio-Logic) with EC-Lab interface. In cyclic voltammetry experiments, the electrodes were cycled at scan rates ranging from 5 to 1000 mV s⁻¹ and data collected were used to understand the kinetics of charging and discharging of the electric double layer. To calculate the specific capacitance (C_s) from CV data the following equation was applied

$$C_s = \frac{1}{m\nu t} \int_0^t I dt \quad (2)$$

where m , ν , I and t represent the total mass of the electrodes (CAG + CF), the CV scan rate, current and total charge/discharge time, respectively. In the galvanostatic studies, a series of specific currents (0.02–3.5 A g⁻¹) were applied and the performance of the symmetric cells were analyzed to extract essential metrics including specific capacitance ($C_{s,cell}$), i.e., capacitance normalized to total mass of the two electrodes (m),

equivalent series resistance (ESR), specific energy (E_s) and specific power (P_s). Eventually, the results were summarized in the form of the Ragone plot. The following equations were applied:

$$C_{s,cell} = \frac{I}{m} \times \left(\frac{dV}{dt} \right)^{-1} \quad (3)$$

$$E_s = \frac{I}{m} \times \int_0^t V(t) dt \quad (4)$$

$$P_s = \frac{E_s}{t} \quad (5)$$

In addition, for symmetric electrode supercapacitors, the specific capacitance of each electrode with mass m_{el} ($C_{s,electrode}$) in the symmetric cell can, in principle, be estimated using Equation (6). Since this

equation unrealistically assumes a perfectly balanced cell with equivalent kinetics at both electrodes, it underestimates the capacitances that would be obtained by a true half-cell measurement. However, it can be useful for comparative purposes.

$$C_{s,electrode} = 4 \times C_{s,cell} \quad (6)$$

The equivalent series resistance (R_{ESR}) was determined from both galvanostatic and potentiostatic electrochemical impedance spectroscopy (EIS). The R_{ESR} can be obtained from the instantaneous voltage drops (ΔV) occurring at the onset of charge and discharge steps [39]:

$$\Delta V = 2 \times I \times R_{ESR} \quad (7)$$

The time constant (τ), $R_{ESR} \times C$, is a measure of how fast the supercapacitor can be charged and discharged, and can be extracted from galvanostatic data by combining Equations (3,7):

$$\tau = R_{ESR} \times m \times C_s = \frac{\Delta V}{2} \times \left(\frac{dV}{dt} \right)^{-1} \quad (8)$$

Impedance measurements were performed at open circuit voltage (OCV) using an AC voltage signal with an amplitude of 5 mV and a frequency range from 200 kHz to 1 mHz. By transforming the EIS data based on the following relationship [40], the complex capacitance (ignoring inductance and parasitic capacitances) can be mathematically represented as:

$$C(\omega) = -\frac{Im Z}{\omega [(Re Z)^2 + (Im Z)^2]} + j \frac{Re Z}{\omega [(Re Z)^2 + (Im Z)^2]} \quad (9)$$

The complex capacitance given by Equation (9) can be resolved into the real and imaginary parts from which two key device performance parameters can be deduced. The real capacitance ($Re C(\omega)$) represents electrostatic charge storage at the electrode-electrolyte interface. Plotting $Re C(\omega)$ as a function of angular frequency, $\omega = 2\pi f$, provides a curve that plateaus at low frequency, from which the capacitance value can be obtained. The imaginary part of the complex capacitance, $Im C(\omega)$, is directly related to electrical resistance, and hence represents energy dissipation in the cell due to resistive losses. The relaxation time constant, at which the device shows predominantly capacitive behavior, can be estimated from the maximum, f_{max} , of the plot of $Im C(\omega)$, versus ω :

$$\tau = \frac{1}{2\pi f_{max}} \quad (10)$$

3. Results and Discussion

3.1. Microstructural characterization

Monolithic CAG-infused spread tow fabrics were successfully prepared by RIFT; the resulting hybrids were rigid lamina with thickness of $107 \pm 4 \mu m$, slightly thicker than the original spread tow fabric (<80

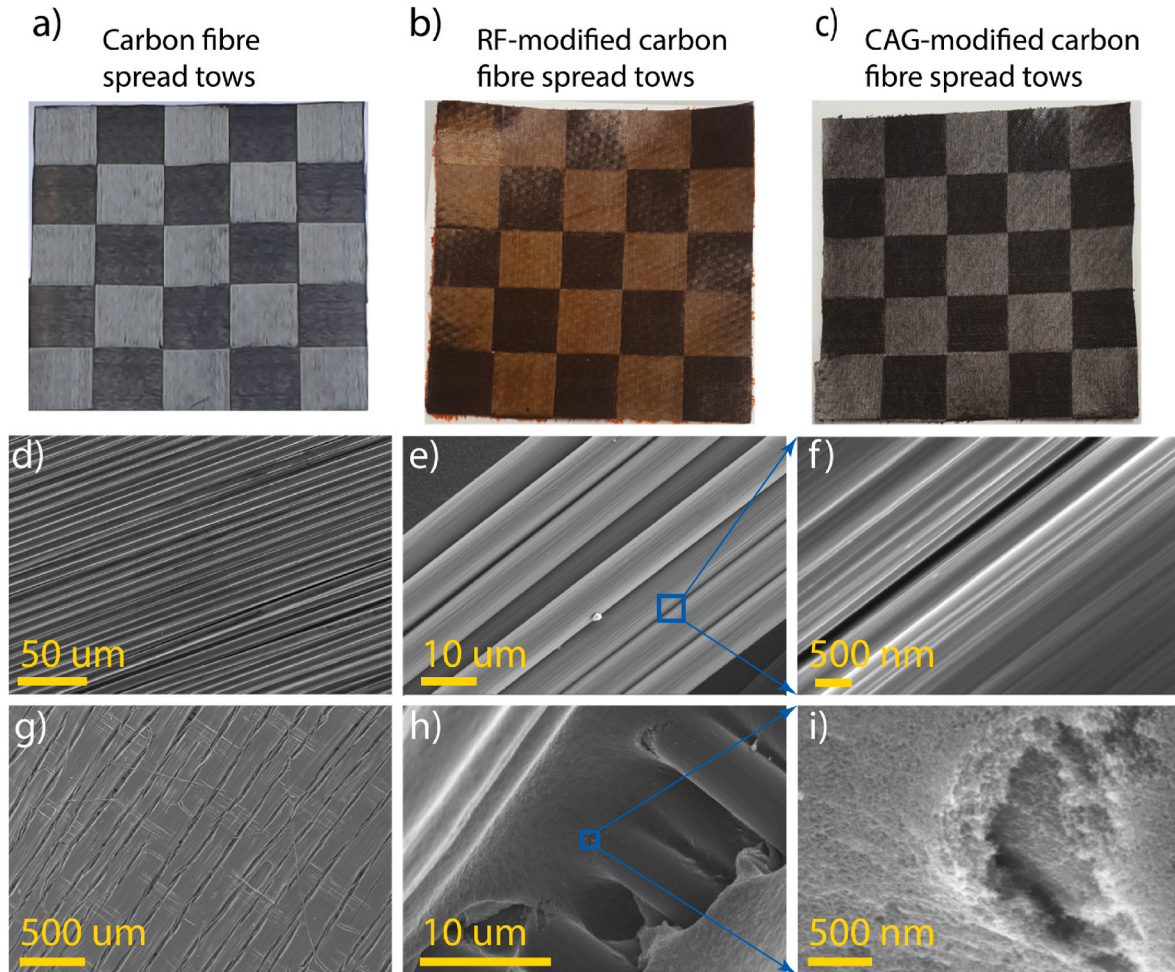


Fig. 2. Photos of TeXtreme® carbon fiber spread tow fabrics (10 cm × 10 cm) before (a) and after modification with resorcinol formaldehyde aerogels (b) and carbon aerogels (c). (d) to (f) display scanning electron micrographs of the as-received fabric (shown left in (a)) at increasing magnification, (g) to (i) display scanning electron micrographs of the fabrics after aerogel pyrolysis (shown right in (c)), at increasing magnification.

μm). Images of the spread tow fabrics (Fig. 2a–c) show a typical warp and weft architecture, with a unit cell of $2 \times 2 \text{ cm}^2$; the RF gel had a red-brown color that converted to black after pyrolysis. The mass of the CAG-CF was 20–30% higher than the CF spread tow treated equivalently, indicating the total relative loading of the carbon aerogel. The TGA curve of the dried precursor RF-infused spread tow CF (see Fig. S4) shows the detail of the carbonisation process. After desorption of additional physisorbed water below 100°C , there is a further weight loss of around 40% associated with breakage of bonds releasing volatile, small molecules during the carbonisation of the RF precursor [41].

The microstructures of active carbon materials, including the size and geometry of pores, and the specific surface areas available for ion adsorption, strongly influence the electrochemical charge storage and power delivery in supercapacitors [42–45]. Hence, the carbon aerogel-modified fabrics were studied using scanning electron microscopy and nitrogen gas physisorption. The scanning electron micrographs (Fig. 2d–f) show the typical surface features of carbon fibers prior to modification with carbon aerogels, which are mainly crenulations running parallel to the fiber axes. The infusion process forms a hybrid microstructure, with carbon aerogel present both on the outer surface of the fabric and within the vacant inter-fiber void volume (Fig. 2g–i). In contrast to the carbon fibres, the carbon aerogel exhibits a highly networked and rigid structure of pores holding the spread tow fibres in place, as desired for both mechanical integrity and electrochemical activity. Part of the CAG in the composite also forms a skin on top of the carbon fibres (Fig. 2g and Figs. S5a–d), as a result of applying RIFT to individual thin plies. The CAG skin exhibits cracks, running parallel to the fibers, at a few hundred micron spacing, which likely originate from the stresses generated during drying, due to pore water evaporation. The cracks expose the interior of the carbon aerogels and the carbon fiber surfaces (Fig. 2g–i and Fig. S5), which in turn may improve accessibility of the surface area to the electrolyte, thereby enhancing the amount of charge stored. Such cracking may potentially serve as mechanically anchoring sites for the multifunctional electrolytes employed in structural supercapacitors, which is the ultimate application of the CAG-CF fabric electrodes [10,46]. The array of cracks could be considered a self-organized version of the coarse holes deliberately introduced in some laminated structural supercapacitors [7], but at a more desirable length scale and orientation.

Gas physisorption analyses showed that modification with carbon aerogel increases the surface area by three orders of magnitude. Based on the isotherms and pore size distributions, the carbon aerogel-carbon fiber samples (Fig. 3) had specific surface areas above $230 \text{ m}^2 \text{ g}^{-1}$ and pore volumes of up to $0.35 \text{ cm}^3 \text{ g}^{-1}$. When these values are normalized only to the mass of the carbon aerogels, the specific surface areas and pore volumes reach $1080 \text{ m}^2 \text{ g}^{-1}$ and $1.6 \text{ cm}^3 \text{ g}^{-1}$, respectively,

comparable to those reported for pure carbon aerogels [35,47–49]. Similar, unmodified carbon fiber fabrics were previously found [10] to have BET specific surface areas of $0.209 \text{ m}^2 \text{ g}^{-1}$ and pore volumes around $0.00029 \text{ cm}^3 \text{ g}^{-1}$. Whilst minor variations are expected with carbon fiber type, it is clear that the addition of aerogel increases BET specific surface area dramatically. In addition, the isotherms (Fig. 3a) show a hysteresis loop with a plateau close to the relative pressure (p/p_0) = 1, which is indicative of capillary condensation that is characteristic of mesoporous materials [37]. This conclusion was consistent with the pore width distribution (Fig. 3b), computed assuming a slit-shaped pore model [38], which indicated that the majority of the pores are smaller than 20 nm (mesopores lying in the range 2–20 nm) and account for most of the cumulative pore volume (Fig. 3c).

3.2. Device fabrication and electrochemical performance testing

To fabricate supercapacitor devices, the carbon fiber-carbon aerogel fabrics were used as freestanding electrodes in a symmetric configuration. Pure electrochemical testing of the supercapacitors was performed in EMI-TFSI ionic liquid electrolyte, which is widely used in commercial supercapacitors. Ionic liquid electrolytes are attractive as they operate in a wider voltage window than aqueous electrolytes and hence an increase in the specific energy density can be achieved, although, the specific power can be limited by the lower ionic conductivity [50]. In the context of structural power systems, the low volatility, and long term stability of ionic liquids is particularly important. Cyclic voltammetry, galvanostatic methods and electrochemical impedance spectroscopy were used to evaluate the electrochemical performance. A two electrode, rather than a three electrode, configuration was selected due to the difficulty of identifying a reliable reference electrode in EMI-TFSI [51], and a desire to evaluate performance in the symmetric configuration intended for eventual structural supercapacitors.

The cyclic voltammograms (CVs) shown in Fig. 4a were recorded at a scan rate of 5 mV s^{-1} and used to evaluate the optimal voltage window for characterizing the supercapacitors using this method. The specific capacitances calculated from the CVs (Fig. 4b) using Equation (2) increased steadily with increasing voltage cut-offs, while increasingly lower coulombic efficiency was observed as the working voltage was extended beyond 2.0 V. This trend is indicative of contributions originating from non-capacitive processes due to oxidative instability of the ionic liquid [50] as a result of charge imbalance between the negative and the positive electrodes [52], for instance, and possibly redox reactions involving trace amounts of water in the electrolyte and oxygen-containing functional groups on the surface of the carbon aerogels [53,54]. These reactions result in a poor coulombic efficiency, at higher voltage, since these Faradaic processes consume electrons that

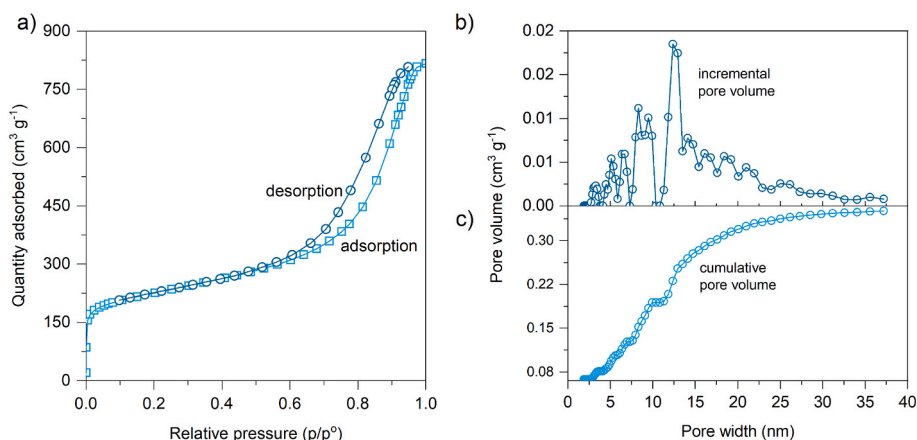


Fig. 3. The specific surface areas and porosity of the carbon aerogel-modified carbon fiber fabrics evaluated using N_2 gas physisorption at 77 K. (a) the isotherms for adsorption and desorption and, (b) the pore width distribution, and (c) the cumulative pore volume.

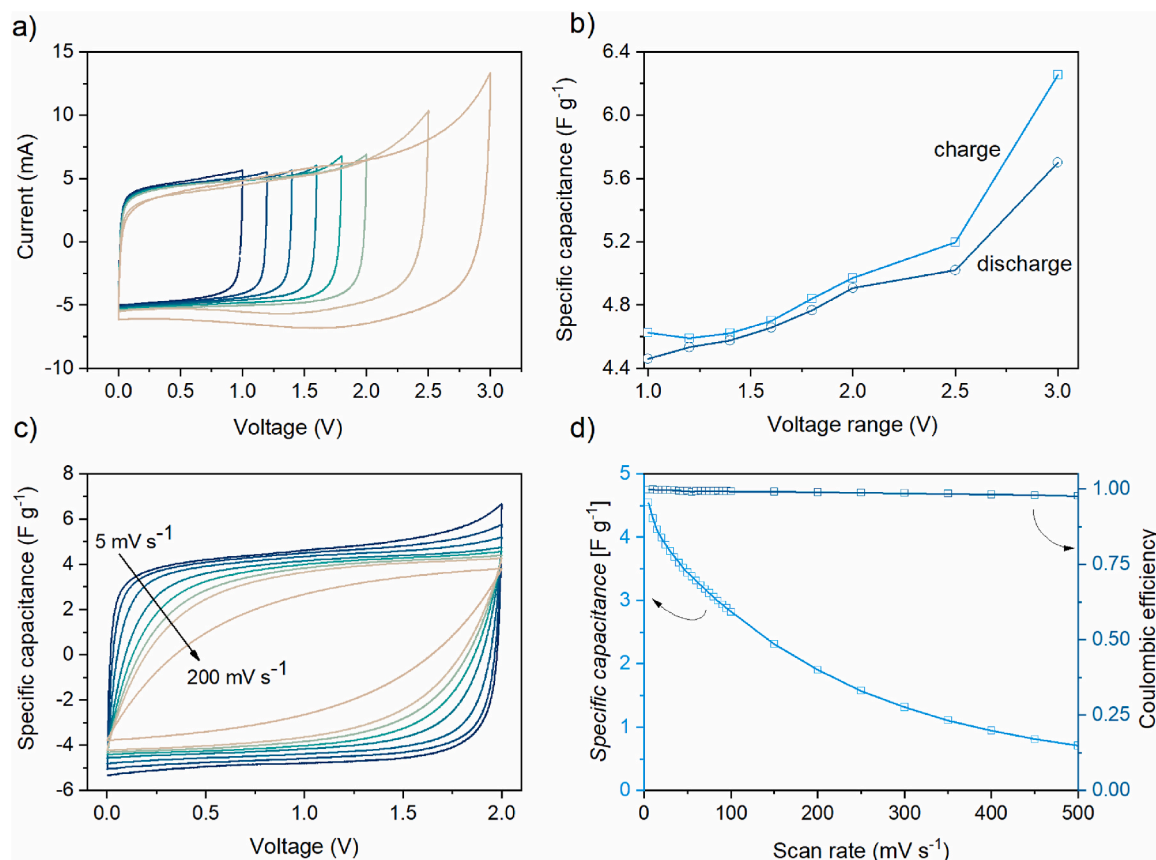


Fig. 4. Electrochemical characterization of symmetric capacitor devices. (a) CV plots of charge and discharge recorded at a scan rate of 5 mV s^{-1} with different increasing cell voltage cut-offs. (b) Specific charge/discharge capacitances deduced from the CV data in (a). (c) CV data obtained at a fixed cell cut-off voltage of 2.0 V for the scan rates $5, 10, 20, 40, 60, 80, 100$ and 200 mV s^{-1} . (d) Specific discharge capacitance and coulombic efficiency derived from (c).

are not recovered during discharge. Further characterizations were performed within the voltage range $0\text{--}2.0 \text{ V}$ at various scan rates to evaluate the supercapacitor performance without significant contributions from Faradaic reactions. The CVs (Fig. 4c and Fig. S6) obtained showed a characteristic capacitive response, with an increasing resistive distortion at higher scan rates. At a scan rate of 5 mV s^{-1} corresponding to full charge and discharge in 400 s , the specific capacitance normalized to the total electrode mass was about 4.5 F g^{-1} . The electrodes maintained a capacitive response, though with reduced capacitance, with increasing scan rate (Fig. 4d) of up to 200 mV s^{-1} ; however, largely resistive behavior was observed in the CV recorded at 1000 mV s^{-1} (Fig. S6), for which the specific capacitance was only 0.24 F g^{-1} , around 5% of the maximum value. These values were normalized to the mass of both electrodes to indicate the maximum CAG-CF electrode performance in a two-electrode symmetric configuration, relevant to a structural supercapacitor. Real specific device performance would, of course, be lower once an optimum structural electrolyte and structural separator were included. On the other hand, the single electrode capacitance of the CAG-CF, inferred from these data (18 F g^{-1} at 5 mV s^{-1}) underestimates the expected 3-electrode performance as the device is not balanced.

Supercapacitors are energy storage devices designed to deliver high power to an external load, and are usually operated (discharged) in either constant current or constant power modes. The ability to retain high specific capacitance during fast charge-discharge process is a crucial requirement for high performance supercapacitors. GCD measurements are therefore useful, in addition to cyclic voltammetry experiments, to assess the electrochemical performance of the CAG-CF symmetric supercapacitors in a more practical context. In the GCD test protocol, prototype symmetric cells were cycled at currents ranging

from 0.02 to 3.5 A g^{-1} . The GCD curves exhibited symmetric and linear voltage-time profiles that are characteristic of capacitors (selected examples in Fig. 5a and Fig. S7). The voltages were allowed to run to 2.7 V , slightly higher than the CV experiments, to allow for IR drops at higher currents; since performance was evaluated from the integrated discharge current, any contributions from adventitious water or electrolyte decomposition are excluded. The galvanostatic data (Fig. 5, Fig. S7 and Fig. S8) were analyzed on the basis of Equations (3–5) to deduce the specific capacitance, energy and power, again normalized to the two active electrodes. For the lowest test current (0.02 A g^{-1}), a maximum discharge capacitance of about 3.1 F g^{-1} was obtained for the symmetric cell which is slightly lower than that obtained from CV test. Whilst the slopes of the curves indicate capacitance, the instantaneous voltage drops measured at the beginning of charge and discharge curves relate to the R_{ESR} . By plotting these voltage drops against current (Fig. 5b), the R_{ESR} (based on Equation (7)) was estimated to be about 3Ω or $48 \Omega \text{ cm}^2$ (normalized to the footprint area of the fabrics). Knowing the R_{ESR} and C values, the time constant was determined using Equation (8) to be about 1.3 s , which is similar to values typically reported for commercial supercapacitors [55]. GCD can also indicate the practical cycling stability of these prototype supercapacitors. In a charge-discharge test at 0.2 A g^{-1} for 3000 cycles (Fig. 5c), the CAG-modified fabric electrodes retained 92% of their initial capacitance (1.8 F g^{-1}) after 1000 cycles, which decreased to 85% at the end of the 3000 th cycle. The capacity fade could be indicative of irreversible parasitic reactions (manifested in coulombic efficiencies $<100\%$ [56]) involving the electrolyte due to charge imbalance, surface functional groups on the carbon aerogels and/or the presence of moisture as cell fabrication was not carried out under dry conditions. This last possibility is partly indicated by coulombic efficiency which was about 97% for the

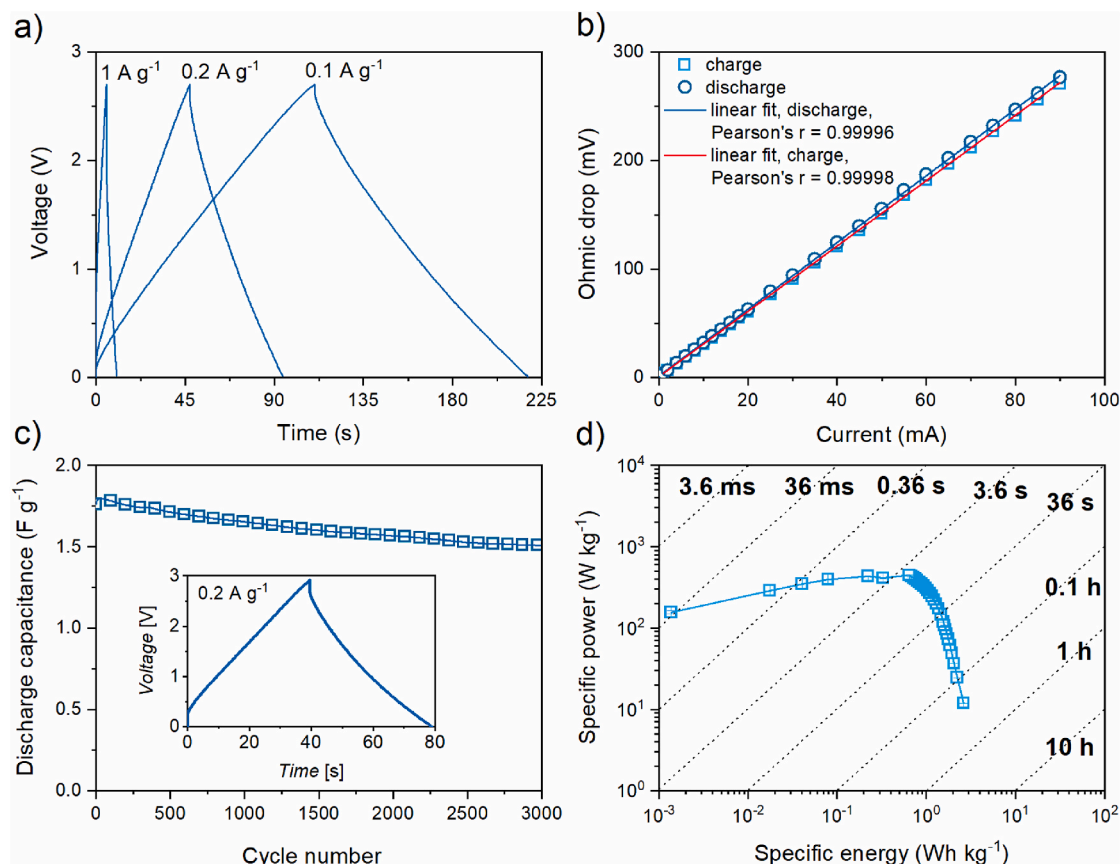


Fig. 5. Galvanostatic charge-discharge (GCD) measurements performed on the symmetric cells in a voltage window ranging from 0 to 2.7 V. A selection of galvanostatic curves for 0.1, 0.2 and 1 A g⁻¹ is shown in (a). (b) The determination of the R_{ESR} from the ohmic drops. (c) A plot showing cycling stability over 3000 cycles at a rate of 0.2 A g⁻¹. (d) The Ragone plot summarizing the specific energies and specific powers for the symmetric cell based on CAG-CF electrodes along with lines representing the response times.

first cycle and stabilized at 99.4% from the 200th cycle onwards as shown in Fig. S9.

Ragone plots provide a performance summary of electrochemical energy storage devices, in terms of specific energy (or energy density) and specific power (or power density). The symmetric cells employing the CAG-CF electrodes achieved maximum specific energies (normalized to the total mass of the two electrodes) and powers of 2.6 Wh kg⁻¹ and 0.44 kW kg⁻¹, respectively (Fig. 5d). The performance data for CAG-CF electrodes fall within the time response regime for materials used in commercial supercapacitors [56]. Supplementary data collected from EIS analysis were used to determine key cell parameters relevant to power delivery. The R_{ESR} determined either from the in-phase plateau of the Bode plot extending up to 200 kHz (Fig. 6a) or the high frequency intercept of the Nyquist plot (Fig. S10), was approximately 2.5 Ω (40 Ω cm²), in good agreement to the value calculated from the GCD data. The maximum of the plot of the imaginary capacitance against the frequency (Fig. 6b) can be used to estimate the time constant, by transforming the EIS data based on Equation (9). Using Equation (10), the value of 2.5 s is on the same order of magnitude as that obtained by GCD, and is within the range typically reported for supercapacitors [50,55,57]. Additionally, the specific capacitance of the cell was estimated from the plot of the real capacitance to be 4.7 F g⁻¹ at 1 mHz, in good agreement to results from CV and GCD tests which were 4.5 and 3.1 F g⁻¹, respectively. Most of the capacitance and energy storage in the hybrid structure is attributed to the carbon aerogel component due to its significantly higher surface area than the carbon fibers. If only the mass of the aerogel in the two electrodes is considered, the specific maximum capacitance is 21 F g⁻¹ with specific energy and power densities of 11.5 Wh kg⁻¹ and 1.9 kW kg⁻¹; these values compare well with other high

surface area carbon systems reported in the literature (Table S1). The specific capacitance of each electrode in the symmetric cell can be estimated using Equation (6), although the approach gives only a lower bound. Normalized to the carbon aerogel mass in a single electrode of CAG-CF fabric, the specific capacitance ranges from 56 to 85 F g⁻¹, as expected for carbon materials characterized in organic and ionic liquid electrolytes for which the capacitance generally ranges from 60 to 150 F g⁻¹ [50,58–64]. This comparison gives a clear indication that the carbon aerogels in the CAG-CF fabric electrodes are utilized efficiently, using the carbon fibres as current collectors (summary data in Table 1). State-of-the-art supercapacitors (manufactured by Maxwell® Technologies, SkelCap® Ultracapacitor, Eaton®, Sech® Capacitors, and IC® Illinois Capacitor) used in consumer electronics have specific capacitance of the whole device ranging from 2 to 6.5 F g⁻¹, maximum specific energy from 2 to 7.8 Wh kg⁻¹, and maximum useable power in the range from 4 to 27 kW kg⁻¹. Assuming an areal mass of 5.4 mg cm⁻² for a 20 μ m thick aluminium foil, which is commonly used in supercapacitors and batteries, the active mass in commercial electrodes is about 48–65 wt% of the total electrode mass, which typically contains 5–10 mg cm⁻² active material [65]. In addition, the active electrode mass in a typical supercapacitor amounts to 30 wt% of the total device mass, with another 15–20 wt% is the metal support and current collector. If the CF-CAG electrode can replace both components (some additional metallization may be needed for current collection, but at much lower mass [66]), the specific capacitance for the two electrodes obtained in this study should be scaled by a factor of 2–3 for comparison. In other words, the performance of the structural electrodes could match the commercial capacitors if the mass loading of the carbon aerogels in the CAG-CF fabrics were increased. However, the total thickness of the

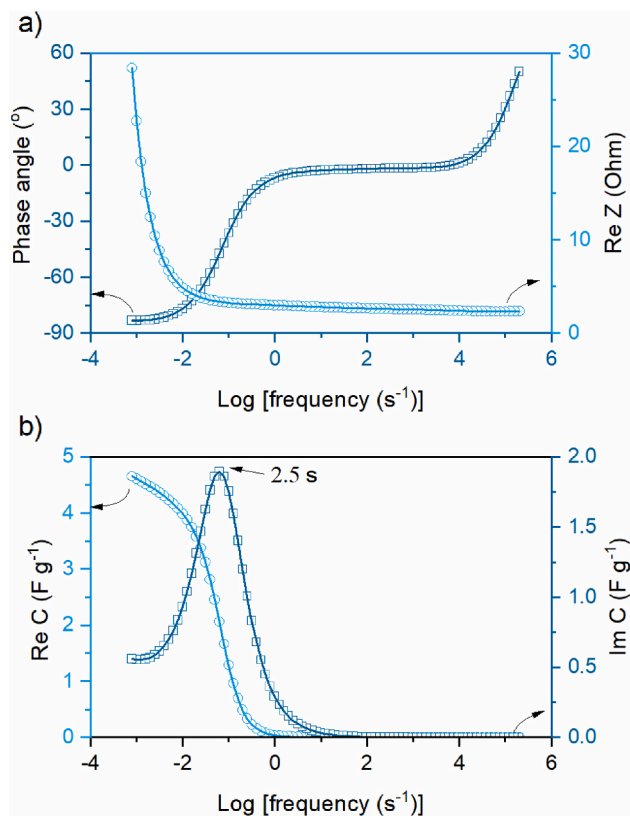


Fig. 6. EIS data for the symmetric cells using two CAG-CF electrodes are presented as a) Bode plots and b) complex capacitance obtained after mathematical transform given in Equation (9).

spread tow plies should be limited, in order to maintain the volume fraction of the reinforcing fibres; hence, a denser aerogel may be desirable. On the other hand, it is not necessary to match the performance of the commercial electrochemical device if the multifunctional system also provides structural function [2]. Overall, the multifunctional role of the electrodes will allow for significant weight-saving and the requirements for energy-storage capacity become less stringent.

As the carbon fibres function as both electrical conductors and mechanical constituents, the CAG-CF fabrics can play a multifunctional role in structural supercapacitors. Notably, the carbon fibres provide the structural support for device fabrication, unlike in conventional devices that rely on the metal foil; the amount of metal required for conductivity is very much lower [66]. The carbon fibres themselves have significant electrical conductivity (on the order of 10^3 S cm^{-1}); whilst not sufficient to provide fully for current collection in larger components, particularly in the case of high-power devices such as supercapacitors. Nevertheless, the contribution of the fibres will significantly reduce the mass of metal required, (partially) replacing the aluminium foil found in conventional supercapacitors. The extent of this substitution will depend on the size of the device and, therefore, the length scale over which current must be conducted. The CAG-CF composite electrodes should provide sufficient

energy and power required for practical applications in structural composite supercapacitor, approaching the performance of commercial monofunctional supercapacitors. However, there is scope to increase the CAG content within the electrodes to improve energy density, and introduce metallisation to improve power density. In any case, the multifunctionality renders the requirements for energy and power less stringent than in conventional supercapacitors in which the electrodes are only energy-storing components. To realise practical structural supercapacitors, exploiting the CAG-CF electrodes, they must be combined with suitable structural electrolytes [67]. Using existing structural electrolytes, previous work showed that CAG-modification of plain weave carbon fiber fabrics improves both electrochemical and mechanical performance [10,46,67]. However, using structural epoxy, the existing CAG-process was found to degrade composite tensile strength significantly, for the equivalent CF/glass fibre ply lay-ups; on the other hand, Young's modulus and the shear modulus were relatively unaffected [6]. This strength degradation may be avoided in the future by process optimisation and fibre selection; thinner plies may be preferred as CAG carbonisation by-products should escape more readily. More generally, thinner plies based on spread tow fiber architectures have shown improved mechanical properties as compared to thicker plies formed from standard carbon fiber fabrics [68]. Therefore, the intrinsically favourable mechanical performance of the spread tow fabrics employed in this study will benefit the multifunctional use of the CAG-CF electrodes in structural supercapacitors. Once the process is optimised, tensile mechanical properties in the fiber direction are expected to reflect the underlying structural fibers used; the spread tow architecture is expected to minimize the tendency for shear-induced delamination. However, one of the greatest challenges in the field remains the preparation of matrices with both sufficient ionic conductivity and mechanical shear strength/stiffness. At present, the use of existing structural electrolytes reduces electrochemical performance, especially power density, significantly. The development of new resin systems, particularly based on bicontinuous nanoarchitectures is an active and promising area of research which will advance the design of better structural supercapacitors [69].

4. Concluding remarks

In this work, RIFT was successfully adapted to infuse resorcinol-formaldehyde (RF) resin, throughout the volume of spread tow carbon fiber fabrics. Polycondensation and ageing resulted in rigid RF aerogel fabrics that converted to carbon aerogels upon heating at 800 °C. Incorporation of the bicontinuous and mechanically stiff carbon aerogels, <30 wt %, into spread tow carbon fiber fabrics increased specific surface area and pore volume very significantly. The BET specific surface area, which is approximately a measure of accessible area in porous electrodes, increased to more than $230 \text{ m}^2 \text{ g}^{-1}$, three orders of magnitude higher than that for plain carbon fibers. As a consequence, the electrochemical capacitance, specific energy and specific power performance of the as-received fabrics were significantly enhanced, approaching practically useful values.

Using EMI-TFSI ionic liquid as an electrolyte extends the working voltage window to 2.7 V, accessing specific energy and power densities which are competitive with state-of-the-art fabric-based

Table 1

A summary of the microstructural and electrochemical characteristics of CAG-modified carbon fiber spread tows used in symmetric cells using EMI-TFSI ionic liquid. Capacitance, Energy and Power are normalized to the total mass of the two electrodes.

Materials	BET specific surface area ($\text{m}^2 \text{ g}^{-1}$)	Pore volume ($\text{cm}^3 \text{ g}^{-1}$)	Specific capacitance (F g^{-1})			Maximum specific energy (Wh kg^{-1}) from GCD	Maximum specific power (kW kg^{-1}) from GCD	R _{ESR} ($\Omega \text{ cm}^2$)		Relaxation time (τ) (s)	
			CV	GCD	EIS			GCD	EIS	GCD	EIS
CAG	1078	1.6	20.5	14.1	21.4	11.5	1.9	48	40	1.3	2.5
CAG-CF spread tow	237	0.35	4.5	3.1	4.7	2.6	0.44				

supercapacitors. The maximum specific capacitance was generally in the range $3\text{--}5\text{ F g}^{-1}$ as determined using GCD, CV and EIS methods; the maximum specific energy and power obtained from GCD measurements were 2.6 W h kg^{-1} and 0.44 kW kg^{-1} , normalized to the total mass of both CAG-CF electrodes. This capacitance can be attributed to the carbon aerogel (23 wt%), all of which is efficiently utilized. The carbon fibers, as well as their direct structural function, serve to support and connect the aerogel monolith, which would otherwise fragment. Once effective structural electrolytes are available, simultaneous electrochemical and mechanical testing will be informative particularly in load cases where the matrix is more significant. Since spread tow carbon fabrics have a third of the thickness of plain weave fabrics and reduced crimp, CAG-modified spread tow fabrics are anticipated to deliver higher electrochemical power densities and perform better mechanically, in complete structural composite supercapacitors. Therefore, the data reported in this paper provide input for future studies on the electro-mechanical performance of these fabric electrodes in a variety of applications.

CRedit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compscitech.2023.110042>.

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